

Module 16: History of Life — Keys to Success

Learning Objectives

By the end of this module, you should be able to:

1. Place major biological milestones in chronological order on the geologic time scale.
2. Explain the relationship between Mass Extinctions and Adaptive Radiations.
3. Correctly read and interpret a phylogenetic tree.
4. Define a "clade" and identify shared derived characters.

The Big Picture

If the 4.6 billion-year history of Earth were compressed into a single 24-hour day, anatomically modern humans wouldn't appear until roughly the last *two seconds* before midnight. Understanding the history of life requires taking off our human-centric glasses and recognizing the massive stretches of "deep time" when bacteria entirely ruled the planet.

Key Terms & Concepts

1. The Geologic Timeline Milestones

You do not need to memorize every era and epoch, but you *must* know these milestones in chronological order:

1. **Earth forms** (~4.6 Billion Years Ago). No life. Toxic atmosphere.
2. **First Life (Prokaryotes)**: (~3.5 BYA). Simple, single-celled bacteria living in oceans. They ruled the earth alone for almost two billion years!
3. **The Oxygen Revolution**: (~2.7 BYA). Photosynthetic bacteria (cyanobacteria) invented a way to use sunlight to make food, releasing toxic oxygen gas as a byproduct. This rusted the iron in the oceans and completely changed the atmosphere.

4. **First Eukaryotes:** (~1.8 BYA). Endosymbiosis occurs! One bacteria swallowed another to form mitochondria (Module 04!). Complex cells arrive.
5. **Multicellular Life:** (~1.2 BYA). Cells begin cooperating to form algae and eventually animals.
6. **The Cambrian Explosion:** (~535 MYA). A massive, rapid diversification of complex animal body plans in the oceans.
7. **Colonization of Land:** (~500 MYA). Plants, fungi, and eventually arthropods (bugs) and tetrapods (amphibians) brave the dry land.

2. Extinctions and Radiations

The history of life is not a smooth ride; it's punctuated by disasters.

- **Mass Extinctions:** There have been 5 major mass extinctions in Earth's history (e.g., the asteroid that wiped out the non-avian dinosaurs 66 MYA). They wipe out >50% of the Earth's species rapidly.
- **Adaptive Radiation:** What happens immediately *after* an extinction? The survivors find themselves in an empty world with no predators and no competition! They rapidly evolve and spread out to fill all the empty ecological niches. (Example: When the dinosaurs died, tiny surviving mammals underwent massive adaptive radiation to become the whales, bats, bears, and primates we see today).

3. Phylogeny and Reading Trees

Phylogeny is the evolutionary history of a species. Scientists draw **Phylogenetic Trees** (or cladograms) to map out who is related to whom.

How to read a tree:

- **The Root:** The base of the tree represents the oldest common ancestor.
- **The Branches:** Time flows from the root up through the branches.
- **The Nodes:** Every time a branch splits into two, there is a "Node." A node represents a *Most Recent Common Ancestor* that lived at that point in time and split into two new, distinct lineages.
- **Sister Taxa:** Two lineages that split from the exact same node. They are each other's closest living relatives.

The Golden Rule of Trees: Do NOT read a tree simply by looking at the tips (the top) from left to right. Relatedness is determined *only* by tracing backwards to see how recently two

organisms shared a common ancestor (a node). If two organisms share a very recent node, they are closely related. If their shared node is deep at the bottom of the tree, they are distantly related.

4. Clades

A **Clade** is a grouping that includes an ancestor and *ALL* of its descendants. - You can spot a true clade on a tree by "cutting" a single branch. Everything that falls off that single cut forms a clade. If you have to make two cuts to isolate a group, the group is not a true clade!

Study Tips

1. **Be careful reading left-to-right.** Trees are like mobiles hanging from the ceiling. You can spin them around any node and they still mean the exact same thing. Relatedness is determined by the depth of the shared node, not by who is sitting next to whom at the top.
2. **Extinction is an opportunity.** Always tie Mass Extinction to Adaptive Radiation. Death paves the way for new life.
3. **Endosymbiosis ties the course together.** Remember how mitochondria and chloroplasts have their own DNA? This module places that event chronologically.